



Python for Data Science - 2305CS303

Lab - 10

Roll No. : 122

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```
import pandas as pd
import numpy as np
```

1. Series in Pandas

Create a Pandas Series containing the marks of 8 students. Display the Series, its index, values, and data type.

```
s= pd.Series([30,40,50,34,45,56,67,45,34,87])
print(s)
```

```
0    30
1    40
2    50
3    34
4    45
5    56
6    67
7    45
8    34
9    87
dtype: int64
```

2. Creating Series from Different Data

Create three Pandas Series: i) from a Python list, ii) from a NumPy array, and iii) from a dictionary. Display all three Series.

```
s= pd.Series([30,40,50,34,45,56,67,45,34,87])
s1=pd.Series(np.array([10,20,30]))
d={'a':10,'b':20}
s2=pd.Series(d)
print("-----list-----")
print(s)
print("-----numpy array-----")
print(s1)
print("-----dictionary-----")
print(s2)
```

```
-----list-----
0    30
1    40
2    50
3    34
4    45
5    56
6    67
7    45
8    34
9    87
```

```
dtype: int64
-----numpy array-----
0    10
1    20
2    30
dtype: int64
-----dictionary-----
a    10
b    20
dtype: int64
```

3. Custom Index in Series

Create a Pandas Series containing the marks of 5 students with student names as the index. Access the marks of two specific students using their labels.

```
stu={'ram':20, 'ravan':30, 'rrr':40, 'ddd':50, 'rt':60}
marks=pd.Series(stu)
print(marks['ram'])
print(marks['ravan'])
```

```
20
30
```

4. Creating Time Series

Create a date range for 10 consecutive days starting from '2026-09-01' and create a Pandas Series containing daily sales values for those dates. Display the resulting time series.

```
dates=pd.date_range(start='2026-09-01', periods=9)
sales=pd.Series([40,50,60,60,20,60,56,12,45], index=dates)
print(sales)
```

```
2026-09-01    40
2026-09-02    50
2026-09-03    60
2026-09-04    60
2026-09-05    20
2026-09-06    60
2026-09-07    56
2026-09-08    12
2026-09-09    45
Freq: D, dtype: int64
```

5. Time Series with Frequency

Create a time series for the first 6 months of 2026 using a suitable date frequency. Store monthly revenue values in the Series and display the result.

```
dates=pd.date_range(start='2026', periods=6, freq='ME')
sales=pd.Series([40,50,60,60,20,60], index=dates)
print(sales)
```

```
2026-01-31    40
2026-02-28    50
2026-03-31    60
2026-04-30    60
2026-05-31    20
2026-06-30    60
Freq: ME, dtype: int64
```

6. Creating a DataFrame

Create a DataFrame named `students` containing the following columns: RollNo, Name, Department, Marks, and City. Add records for at least 6 students. Display the DataFrame.

```
d={'rollno':[101,102,103,104,105,106],
  'Name':['raj','rahul','ram','ravan','rohan','ritik'],
  'Department':['CE','CE','MANAGEMENT','MANAGEMENT','CE','MANAGEMENT'],
  'Marks':[30,40,50,60,70,80],
  'City':['rajkot','raipur','raigadh','rampur','rome','ranchi']}
student=pd.DataFrame(d)
print(student)
```

	rollno	Name	Department	Marks	City
0	101	raj	CE	30	rajkot
1	102	rahul	CE	40	raipur
2	103	ram	MANAGEMENT	50	raigadh
3	104	ravan	MANAGEMENT	60	rampur
4	105	rohan	CE	70	rome
5	106	ritik	MANAGEMENT	80	ranchi

7. DataFrame from Dictionary

Create a DataFrame from a dictionary containing information about 5 products: ProductID, ProductName, Category, Price, and Quantity. Display the DataFrame and its columns.

```
p={'ProductID':[101,102,103,104,105,106],
  'ProductName':['raj','rahul','ram','ravan','rohan','ritik'],
  'Category':['CE','CE','MANAGEMENT','MANAGEMENT','CE','MANAGEMENT'],
  'Price':[300,400,500,600,700,800],
  'Quantity':[10,20,201,305,604,802]}
product=pd.DataFrame(p)
print(product)
```

	ProductID	ProductName	Category	Price	Quantity
0	101	raj	CE	300	10
1	102	rahul	CE	400	20
2	103	ram	MANAGEMENT	500	201
3	104	ravan	MANAGEMENT	600	305
4	105	rohan	CE	700	604
5	106	ritik	MANAGEMENT	800	802

8. Conditional Selection

Using the `students` DataFrame, display: i) students having Marks greater than 70, ii) students belonging to the 'CSE' department, and iii) students having Marks between 60 and 80.

```
print("Marks greater than 70:-\n",student[student['Marks']>70])
print("Belong to CE:-\n",student[student['Department']=='CE'])
print("Marks between 60 and 80:-\n",student[(student['Marks']>60) & (student['Marks']<80)])
```

```
Marks greater than 70:-
  rollno  Name  Department  Marks  City
5    106  ritik  MANAGEMENT    80  ranchi
Belong to CE:-
  rollno  Name  Department  Marks  City
0    101   raj         CE     30  rajkot
1    102  rahul         CE     40  raipur
4    105  rohan         CE     70   rome
Marks between 60 and 80:-
  rollno  Name  Department  Marks  City
4    105  rohan         CE     70   rome
```

9. Multiple Conditions

Using the `students` DataFrame, select students who belong to the 'CSE' department and have Marks greater than or equal to 70. Also display students who belong to either 'CSE' or 'IT'.

```
print("Marks between 60 and 80:-\n",student[(student['Department']=='CE') & (student['Marks']>=70)])
```

```
Marks between 60 and 80:-
  rollno  Name  Department  Marks  City
4    105  rohan         CE     70   rome
```

10. Conditional Selection on DataFrame Columns

From the `products` DataFrame, display products whose Price is greater than 500 and Quantity is less than 20. Then display only the ProductName and Price columns of the selected records.

```
print(product[(product['Quantity']<20) & (product['Price']>=200)][['ProductName','Price']])
```

	ProductName	Price
0	raj	300

11. Read CSV File

Create a CSV file named `students.csv` containing student records. Read the CSV file using Pandas and display the complete DataFrame. Also display its first 5 records.

```
url = 'https://raw.githubusercontent.com/ShapeLab/ZooidsCompositePhysicalizations/master/Zooid_Vis/bin/data/student-dataset'
student = pd.read_csv(url)
print(student.head())
```

id	name	nationality	city
0	Kiana Lor	China	Suzhou
1	Joshua Lonaker	United States of America	Santa Clarita
2	Dakota Blanco	United States of America	Oakland
3	Natasha Yarusso	United States of America	Castro Valley
4	Brooke Cazares	Brazil	São José dos Campos

	latitude	longitude	gender	ethnic.group	age	english.grade	math.grade
0	31.31	120.62	F	NaN	22	3.5	3.7
1	34.39	-118.54	M	NaN	22	2.9	3.2
2	37.80	-122.27	F	NaN	22	3.9	3.8
3	37.69	-122.09	F	NaN	20	3.3	2.8
4	-23.18	-45.88	F	NaN	21	3.7	2.6

	sciences.grade	language.grade	portfolio.rating	coverletter.rating
0	3.1	1.0	4	4.0
1	3.6	5.0	5	4.0
2	3.2	5.0	3	3.0
3	3.2	5.0	5	2.0
4	3.4	1.0	4	4.0

	refletter.rating
0	4
1	5
2	4
3	4
4	5

12. Read Excel File

Create an Excel file named `students.xlsx` containing student records. Read the Excel file using Pandas and display the DataFrame. Display the column names and shape of the DataFrame.

```
# datafile:- https://github.com/SanyaShresta25/student-registration-system/blob/main/Student_data.xlsx
url = 'Student_data.xlsx'
student = pd.read_excel(url)
print(student)
```

	Registration No.	Name	Class	Gender	DOB
0	1	Sanya Shresta	5	2	25/04/2004
1	2	Saanvi U.	5	1	02/03/2004
2	3	Laila	6	1	01/01/2003
3	4	Amila	6	1	23/04/2007
4	5	Bravo Mitchell	10	1	23/04/1999
5	6	Bella	8	1	23/05/2004
6	7	Nidhi	8	2	02/02/2004
7	8	Meena	9	2	23/09/2004
8	9	Pinky	4	1	12/09/2004
9	10	Sayeem	4	1	12/03/2004
10	11	Nishan	9	1	23/09/2006

	Date of Registration	Religion	Skill	Father Name	Mother Name
0	16/04/2024	Christian	Python	Deeshan	Deeksha
1	16/04/2024	Hindu	Java	Niranjan	Naina
2	16/04/2024	Muslim	C	Ahad	Insha
3	16/04/2024	Hindu	Python Dev	Kishan	Smrithi
4	16/04/2024	Christian	Cricketer	Gary	Lena
5	16/04/2024	Hindu	C	Codd	Anna
6	17/04/2024	Hindu	Python	Papa	Amma
7	17/04/2024	Hindu	Python	deeshan	deeksha
8	17/04/2024	Hindu	Node.js	Kushagr	No
9	17/04/2024	Muslim	Everything	abba	ammi
10	18/04/2024	Christian	mySQL	Vedang	Hema

	Father's Occupation	Mother's Occupation
0	Engineer	Homemaker
1	Financial Adviser	Homemaker
2	SDE	SDE
3	Cricketer	Lecturer
4	TV reporter	Singer
5	SDE	SdE
6	Kelsa	Kelsa
7	Engineer	homemaker
8	Vela	mafia
9	sde	sde
10	ML specialist	Homemaker

13. Setting Index

Using the `students` DataFrame, set the `RollNo` column as the index. Display the DataFrame and access the record of a student using the index label.

```
student=student.set_index("Registration No.")
print(student.loc[2])
```

```
level_0          1
index            1
Name            Saanvi U.
Class            5
Gender            1
DOB            02/03/2004
Date of Registration  16/04/2024
Religion          Hindu
Skill             Java
Father Name       Niranjan
Mother Name       Naina
Father's Occupation  Financial Adviser
Mother's Occupation  Homemaker
Name: 2, dtype: object
```

14. Resetting Index

Reset the index of the DataFrame created in the previous question. Demonstrate both keeping the old index as a column and removing the old index.

```
student=student.reset_index()
```

15. Integrated Pandas Exercise

Create a student DataFrame containing RollNo, Name, Department, Marks, and City. Set RollNo as the index, select students with Marks >= 75, reset the index, and finally save the resulting DataFrame as both CSV and Excel files.

```
student_data={'rollno':[1,2,3,4],
              'name':['yatharth','sahil','vimal','akshay'],
              'department':['CE','MECH','CE','MECH'],
              'marks':[12,34,56,78],
              'city':['rajkot','jamnagar','rajkot','dhandhuka']}
student=pd.DataFrame(student_data)
student=student.set_index('rollno')
res=student[student['marks']>=75]
print("marks greater than 75\n",res)
student=student.reset_index()
res.to_csv("top_student.csv")
res.to_excel("top_student.xlsx")
```

```
marks greater than 75
   name department  marks    city
rollno
4    akshay      MECH    78 dhandhuka
```